

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim$$

LINEAR ALGEBRA I

Vectors and linear combinations

Course orientation • 80-minute exposition • 90-minute worked examples

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Wednesday cohorts • Room A-207

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Today's three blocks

Course framework and working procedures

Vectors as column objects

Addition and scalar multiplication

Linear combinations and their geometry

Problems followed immediately by solutions



Wednesday teaching blocks

Morning cohort

- BS26-1–5 • 66 students
- 09:00–10:20 exposition
- 10:30–12:00 examples
- Room A-207

Afternoon cohort

- BS26-6–10 • 70 students
- 13:00–14:20 exposition
- 14:30–15:50 examples
- Room A-207



Course structure

15 teaching weeks

6 ECTS and 180 hours total workload

60 lecture hours and 30 practical hours

90 hours of independent study



Course progression

Weeks 1–4 vectors and matrices

Weeks 5–10 systems and vector spaces

Weeks 11–12 orthogonality and least squares

Weeks 14–15 eigenvalues and SVD



Assessment and grading

Assessment

- Test 1 15%
- Midterm 30%
- Test 2 15%
- Final examination 40%

Course grade

- A 90–100
- B 75–89
- C 60–74
- D 0–59



Attendance and Moodle

Attend at least 75% of meetings

Attendance is recorded for lectures and practical classes

Weekly materials are posted in Moodle

Official absence documentation is handled administratively



Responsible use of AI

Useful for learning

- Clarifying a definition
- Generating extra practice
- Checking an approach
- Debugging a calculation

Not a substitute for your work

- Do not submit generated work as your own
- Acknowledge and describe permitted AI help
- Be prepared to explain every step



How each week will work

Understand the central idea

See a geometric or applied example

Solve problems individually and in groups

Explain the reasoning aloud

Continue with independent study

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim$$

Part I The heart of the subject

Add vectors. Multiply vectors by numbers. Combine those two operations.



Two operations generate the subject

Start with vectors v and w and scalars c and d

$$v + w$$

$$cv$$

$$cv + dw$$

The expression $cv + dw$ is a linear combination



A vector keeps quantities together

Column form

- $v = [v_1; v_2]$
- v_1 is the first component
- v_2 is the second component

Geometric form

- Arrow from the origin
- Endpoint (v_1, v_2)
- The same vector, three representations



Add components that belong together

For $v=(v_1,v_2)$ and $w=(w_1,w_2)$

$$\mathbf{v} + \mathbf{w} = (v_1+w_1, v_2+w_2)$$

First component with first; second with second



Scale every component

For a scalar c and vector $v=(v_1, v_2)$

$$cv = (cv_1, cv_2)$$

Positive c stretches; negative c reverses; $c=0$ gives 0



Four special combinations

Combination

- $v+w$ sum
- $v-w$ difference
- $0v+0w$ zero vector
- $cv+0w$ multiple of v

Geometric meaning

- Parallelogram diagonal
- Reverse one arrow
- Return to the origin
- Stay on the line through v



Example Addition and subtraction

Let $v=(4,2)$ and $w=(-1,2)$

$$\mathbf{v+w=(3,4)} \quad \mathbf{v-w=(5,0)}$$

The two results are the diagonals formed from v and w



Vectors extend beyond the plane

Two components give a point in $\mathbb{R}^2$

Three components give a point in $\mathbb{R}^3$

The same component rules work in $\mathbb{R}^n$

The picture survives even when drawing becomes impossible



A typical combination in $\mathbb{R}^3$

For vectors u , v and w

$$cu + dv + ew$$

Choose three scalars, multiply, then add component by component

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim$$

From one combination to all combinations

The decisive question is what all choices of the coefficients can produce.



One vector gives a line

All multiples

- cu for every real c
- $c=0$ includes the origin
- Positive and negative directions

Special cases

- $u=0$ gives only the zero vector
- Restricting $c \geq 0$ gives a half-line
- Integer c gives separated points



Two vectors usually give a plane

If v is not parallel to u

- cu gives one line
- dv gives a second direction
- $cu+dv$ fills their plane

If v is parallel to u

- v adds no new direction
- Every combination stays on one line
- The second vector is redundant



Three vectors usually give space

If w is outside the uv -plane

- $cu+dv$ fills a plane
- ew moves through parallel planes
- All combinations fill $\mathbb{R}^3$

If w lies in the uv -plane

- w is already a combination of u and v
- No new direction appears
- All combinations remain in the plane



The formal word is span

Collect every possible linear combination

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{\mathbf{c}_1\mathbf{v}_1 + \dots + \mathbf{c}_k\mathbf{v}_k\}$$

Span answers: what vectors can these columns produce?



The next central equation

Put the vectors into the columns of A

$$Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n = b$$

Solving $Ax=b$ asks for weights that produce b



Independent vectors add directions

Independent

- Each new vector adds information
- No vector is built from the others
- Only zero coefficients produce zero

Dependent

- At least one vector is redundant
- A nonzero combination produces zero
- The span is unchanged after removing redundancy



Independence as an equation

Test a set $v_1, \dots, v_k$ by asking

$$c_1 v_1 + \dots + c_k v_k = 0$$

Only the all-zero solution means independence



End of the exposition

Vectors are column objects with components

Addition and scaling are componentwise

Linear combinations are the central operation

All combinations typically form line, plane, space

Independence detects whether a direction is new

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



Part II Problems and solutions

Attempt each problem before revealing the next slide.



Problem 1 Addition and geometry

Let $v=(3,1)$ and $w=(-2,4)$.

Compute $v+w$ and $v-w$.

Draw v , w and $v+w$ head-to-tail.

Which diagonal of the parallelogram represents $v-w$?

Work first. The solution is on the next slide.



Solution 1 Addition and geometry

$$\mathbf{v} + \mathbf{w} = (1, 5) \quad \mathbf{v} - \mathbf{w} = (5, -3)$$

1. Add or subtract matching components.
2. For $\mathbf{v} + \mathbf{w}$, place the tail of $\mathbf{w}$ at the head of $\mathbf{v}$.
3. For $\mathbf{v} - \mathbf{w}$, reverse $\mathbf{w}$ and add; equivalently it connects the endpoint of $\mathbf{w}$ to the endpoint of $\mathbf{v}$.



Problem 2 Recover two vectors

Suppose $\mathbf{v} + \mathbf{w} = (6, 2)$ and $\mathbf{v} - \mathbf{w} = (2, 4)$.

Find $\mathbf{v}$ and $\mathbf{w}$.

Check both given equations.

Explain why adding the two equations is useful.

Work first. The solution is on the next slide.



Solution 2 Recover two vectors

$$2\mathbf{v}=(8,6) \Rightarrow \mathbf{v}=(4,3) \quad 2\mathbf{w}=(4,-2) \Rightarrow \mathbf{w}=(2,-1)$$

1. Add the equations to eliminate w .
2. Subtract the second equation from the first to isolate w .
3. Verify: $\mathbf{v}+\mathbf{w}=(6,2)$, $\mathbf{v}-\mathbf{w}=(2,4)$.



Problem 3 A particular combination

Let $v=(2,-1,1)$ and $w=(0,3,-2)$.

Compute $2v-w$.

Write the general combination $cv+dw$.

Which component equations give $cv+dw=(4,1,0)$?

Work first. The solution is on the next slide.



Solution 3 A particular combination

$$2\mathbf{v}-\mathbf{w}=(4,-5,4) \quad c\mathbf{v}+d\mathbf{w}=(2c,-c+3d,c-2d)$$

1. Multiply each vector by its scalar.
2. Add component by component.
3. For target $(4,1,0)$: $2c=4$, $-c+3d=1$, $c-2d=0$.
4. The equations give $c=2$, $d=1$ and $c\mathbf{v}+d\mathbf{w}=(4,1,0)$.



Problem 4 Line or plane

Describe all combinations of $u=(1,2)$ and $v=(-2,-4)$.

Do they fill a line or the plane?

Find a vector in $\mathbb{R}^2$ that cannot be produced.

Give a nonzero combination that equals zero.

Work first. The solution is on the next slide.



Solution 4 Line or plane

$$v = -2u \quad \text{so} \quad cu + dv = (c - 2d)u$$

1. Every combination is a multiple of u .
2. The span is one line through the origin.
3. For example, $(0,1)$ is not on that line.
4. The relation $2u+v=0$ proves dependence.



Problem 5 A plane in $\mathbb{R}^3$

Let $v=(1,1,0)$ and $w=(0,1,2)$.

Write the general combination $cv+dw$.

Find a simple equation satisfied by every combination.

Is $b=(2,5,6)$ in the plane?

Work first. The solution is on the next slide.



Solution 5 A plane in $\mathbb{R}^3$

$$cv+dw=(c,c+d,2d) \quad \text{therefore } 2y=2x+z$$

1. The first component gives $x=c$ and the third gives $z=2d$.
2. Then $y=c+d=x+z/2$.
3. For $b=(2,5,6)$, the condition gives $10=4+6$.
4. Thus b is in the plane, with $c=2$ and $d=3$.



Problem 6 Weights that produce a target

Find c and d so that

$$c(2, -1) + d(-1, 3) = (5, 0).$$

Write the two scalar equations.

Interpret c and d as weights of the two columns.

Work first. The solution is on the next slide.



Solution 6 Weights that produce a target

$$2c - d = 5 \quad -c + 3d = 0 \quad \Rightarrow \quad c = 3, d = 1$$

1. Equating components gives two equations.
2. The second gives $c = 3d$.
3. Substitution in the first gives $5d = 5$.
4. Check: $3(2, -1) + (-1, 3) = (5, 0)$.



Problem 7 Does a third vector add a direction?

$u=(1,0,1)$, $v=(0,1,1)$, $w=(2,1,3)$.

Is w in $\text{span}\{u,v\}$?

Do u,v,w fill a plane or all of $\mathbb{R}^3$?

Find a dependence relation.

Work first. The solution is on the next slide.



Solution 7 Does a third vector add a direction?

$$w=2u+v \quad \Rightarrow \quad 2u+v-w=0$$

1. The coefficients $c=2$ and $d=1$ reproduce w .
2. Therefore w lies in the uv -plane.
3. It adds no new direction, so the three vectors remain in a plane.
4. The nonzero relation proves dependence.



Problem 8 Restricted combinations

Let $i=(1,0)$ and $j=(0,1)$.

Describe ci for integer c .

Describe $ci+dj$ when $c \geq 0$ and d is any real number.

How does this differ from $\text{span}\{i,j\}$?

Work first. The solution is on the next slide.



Solution 8 Restricted combinations

$$ci+dj=(c,d)$$

1. Integer c gives separated points on the x -axis.
2. With $c \geq 0$ and arbitrary d , the combinations fill the right half-plane.
3. The span allows all real c and d , so it fills all $\mathbb{R}^2$.
4. Restrictions on coefficients can produce sets that are not vector spaces.



Main ideas from Week 1

The arithmetic is componentwise

The geometry comes from all coefficient choices

One direction gives a line; two new directions give a plane

$Ax=b$ is a question about combinations of columns

Dependence means that a vector adds no new direction